Numerical analysis / Numerical Methods

- In reality "Solving a math problem" generally involves finding *an* answer rather than *exact* answer.
- **Numerical analysis** is the study of algorithms that use numerical approximation for the problems of mathematical analysis.
- A numerical method is a complete and definite set of procedures for the solution of a problem, together with computable error estimates. The study and implementation of such methods is the field of numerical analysis/ numerical methods:
- A trick that lets you get closer and closer to an exact answer is a "numerical method".
- Numerical methods find solutions *close* to the answer without ever knowing what that answer is. As such, an important part of every numerical method is a proof that it works.

Analytic method exist b

data/complex functions

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Lets take the example of 4th degree polynomials (Quartic equation)

$$f(x) = ax^4 + bx^3 + cx^2 + dx + e,$$ 4 roots

$$x_{1,2} = -\frac{b}{4a} - S \pm \frac{1}{2}\sqrt{-4S^2 - 2p + \frac{q}{S}}$$

$$x_{3,4} = -\frac{b}{4a} + S \pm \frac{1}{2}\sqrt{-4S^2 - 2p - \frac{q}{S}}$$

where p and q are the coefficients of the second and of the first degree respectively

$$p = \frac{8ac - 3b^2}{8a^2}$$

$$S = \frac{1}{2}\sqrt{-\frac{2}{3}p + \frac{1}{3a}\left(Q + \frac{\Delta_0}{Q}\right)}$$

$$q = \frac{b^3 - 4abc + 8a^2 d}{8a^3}$$

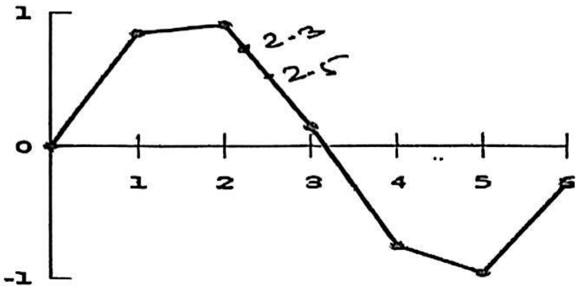
$$Q = \sqrt[3]{\frac{\Delta_1 + \sqrt{\Delta_1^2 - 4\Delta_0^3}}{2}}$$

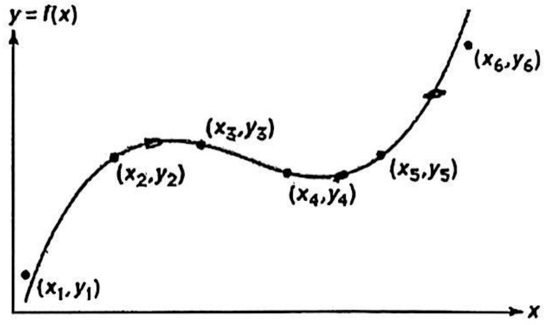
$$\Delta_0 = c^2 - 3bd + 12ae$$

$$\Delta_1 = 2c^3 - 9bcd + 27b^2 e + 27ad^2 - 72ace$$

Data available does not admit applicability of direct analytic method

- Interpolation :- 2001 2011 2009





Applications

- The formal academic area of numerical analysis ranges from quite theoretical mathematical studies to computer science issues.
- Such problems arise throughout the natural sciences, social sciences, engineering, medicine, and business.
- The growth in computing power has revolutionized the use of realistic mathematical models in science and engineering, and subtle numerical analysis is required to implement these detailed models of the world. For example, ordinary differential equations appear in celestial mechanics (predicting the motions of planets, stars and galaxies); numerical linear algebra is important for data analysis; in simulating living cells for medicine and biology.

Applications

Usually used in computer science for root algorithm.

- It is used to determine profit and loss in the company.
- Used for Multidimensional root finding.
- Solving practical technical problems using scientific and mathematical tools
- Network Simulation
- Train and Traffic signal
- Weather prediction
- Build up a algorithm

NUMERICAL METHODS

Accuracy of Numbers

We have two types

(1) Exact numbers

The numbers of the type $8,1,\frac{1}{2},\frac{3}{4},9.75$ will be in category of exact numbers .

(2) Approximate numbers

The numbers $\frac{1}{3}=0.333333....,\ \pi=3.141592.....$ do not have finite decimal expansion and hence they are approximated to some finite digits called as significant digits for the purpose of calculations.

Error in Numerical computation

We have following types of errors in numerical calculations :

(i)Inherent Errors : Errors which exist in the problem either due to approximate given data limitations of computing aids ,are called as inherent errors .The inherent error can be minimised by taking better data and using high precision computing aids

(ii)Round-off Errors : These errors are due to rounding off the numbers while process of computations. The round off errors can be reduced by doing the computations to more significant digits than given in data at each stage of computation.

Ex. Some times we have numbers with a large number of digits such as 7.5846712 which can be rounded off to a significant figure 7.585. This process is called as rounding off the numbers.

$$\Rightarrow 7.5846712-7.585=-0.0003288 \text{ (Round-off error)}$$

(iii)Truncation Errors : These errors arise due to use of approximate formula in computation or by truncating the infinite series to some approximate terms. The study of this type of error is usually associated with the problem of convergence of infinite series.

Ex. $y=y(x)=y(0)+xy'(0)+\frac{x^2}{2!}y''(0)+\frac{x^3}{3!}y'''(0)+...+\frac{x^n}{n!}y^{(n)}(0)+...$

let $y=1+\frac{x}{2}+\frac{x^2}{2}+\frac{3x^3}{8}+.....$

$\therefore\ y(0.1)=1+\frac{0.1}{2}+\frac{(0.1)^2}{2}+\frac{3(0.1)^3}{8}+.....$

$\therefore\ y(0.1)\approx 1.055375\approx 1.0554$

$(0.1)^2=0.01$

$(0.1)^3=0.001$

$(0.1)^4=0.0001\approx 0$

$(0.1)^5=0.00001\approx 0$

(iv) Absolute, Relative and Perecentage Errors : If X is true value of quantity and X' is its approximate value, then

(a) Absolute error $E_A = |X - X'|$. $\Rightarrow |7.5846712 - 7.585| = |-0.0003288| = 0.0003288$

(b) Relative error $E_R = \left|\frac{X - X'}{X}\right|$

$$\Rightarrow \left|\frac{7.5846712 - 7.585}{7.5846712}\right| = |-0.00004335059376| = 0.00004335059376$$

(c) Percentage error $E_P = \left|\frac{X - X'}{X}\right| \times 100$

$$\Rightarrow \left|\frac{7.5846712 - 7.585}{7.5846712} \times 100\right| = |-0.00004335059376| \times 100 = 0.004335059376$$

If ΔX be a number such that $|X - X'| \le \Delta X$, then ΔX is an upper limit of the magnitude of absolute error and measures the absolute accuracy.

$\frac{\Delta X}{|X|} \approx \frac{\Delta X}{|X'|}$ measures the relative accuracy.

NOTE : If two numbers are added or substracted, then magnitude of the absolute error in the result is the sum of the magnitudes of the absolute errors in the two numbers.

General Formula for Error

Let $u = f(x, y, z)$ be the function of 3 variables.

$$\therefore \Delta u = \frac{\partial f}{\partial x}\Delta x + \frac{\partial f}{\partial y}\Delta y + \frac{\partial f}{\partial z}\Delta z$$

Let $u = f(x_1, x_2, ---x_n)$ be the function of n variables.

$$\therefore \Delta u = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\Delta x_i = \frac{\partial f}{\partial x_1}\Delta x_1 + \frac{\partial f}{\partial x_2}\Delta x_2 + ---- + \frac{\partial f}{\partial x_n}\Delta x_n.$$

Then relative error E_R in u is given by,

$$E_R = \frac{\Delta u}{u}$$

Ex.1: Find relative maximum error in the function

$$u = \frac{5xy^2}{z^3} \text{ at } x = y = z = 1 \text{ with } \Delta x = \Delta y = \Delta z = 0.001.$$

Solution : We have $u = \dfrac{5xy^2}{z^3}$,

$$\therefore \frac{\partial u}{\partial x} = \frac{5y^2}{z^3}, \quad \frac{\partial u}{\partial y} = \frac{10xy}{z^3} \quad \text{and} \quad \frac{\partial u}{\partial z} = -\frac{15xy^2}{z^4}$$

$$\frac{\partial}{\partial z}\left(\frac{1}{z^3}\right) = \frac{\partial}{\partial z}(z^{-3}) = -3z^{-4} = \frac{-3}{z^4}$$

$$\therefore \Delta u = \frac{\partial u}{\partial x}\Delta x + \frac{\partial u}{\partial y}\Delta y + \frac{\partial u}{\partial z}\Delta z$$

$$\therefore \Delta u = \frac{5y^2}{z^3}\Delta x + \frac{10xy}{z^3}\Delta y - \frac{15xy}{z^4}\Delta z$$

$$\therefore (\Delta u)_{\max} = \left|\frac{5y^2}{z^3}\Delta x\right| + \left|\frac{10xy}{z^3}\Delta y\right| + \left|-\frac{15xy^2}{z^4}\Delta z\right|$$

$$\therefore (\Delta u)_{\max} = 5(0.001) + 10(0.001) + 15(0.0001) = 0.03$$

$$(E_R)_{\max} = \frac{(\Delta u)_{\max}}{u} = \frac{0.03}{5} = 0.006.$$

(a) Algebraic equation of n^{th} degree has the form:

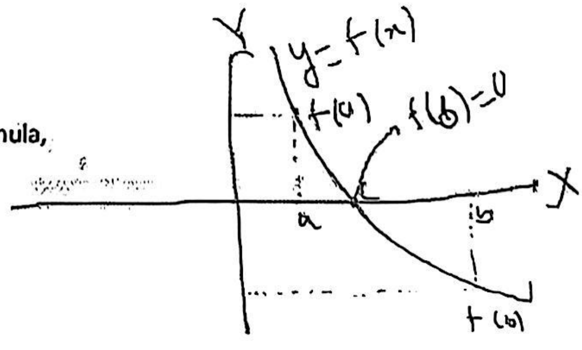
$$f(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_2 x^2 + a_1 x + a_0 = 0 \quad \text{------------------(1)}$$

Ex. (i) $x^4 - x^3 + 2x^3 - 10 = 0$ (ii) $x^3 + 2x^2 + 3 = 0$ (iii) $x^2 - 5x + 6 = 0$

Consider, $ax^2 + bx + c = 0$

Roots of the quadratic equations can be found by the formula,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



For n^{th} degree algebraic equation,

(i) $f(x) = 0$ has exactly n roots, real or complex.

(ii) If $f(x)$ is exactly divisible by $x - a$, then $x = a$ is a root of $f(x) = 0$ i.e $f(a) = 0$

(iii) If $f(a)$ and $f(b)$ have opposite signs then, $f(x) = 0$ has at least one root lying between a and b

(b) Transcendental equation: Non algebraic equation is called as transcendental equation.

A transcendental equation is an equation containing transcendental functions. Elementary transcendental functions are the exponential, logarithmic, trigonometric, reverse trigonometric, and hyperbolic functions.

Ex. (i) $f(x)=\sin x-3x+1=0$ (ii) $f(x)=xe^{x}-2=0$ (iii) $f(x)=x+\log_{10}x-12=0$

Following methods are used to find the roots of algebraic and transcendental equations:

(1) Newton-Raphson method (for non- repeated roots),

(2) Generalised Newton's method for equal roots,

(3) Method of False position(Regula-false method)

(4) Direct Iteration method

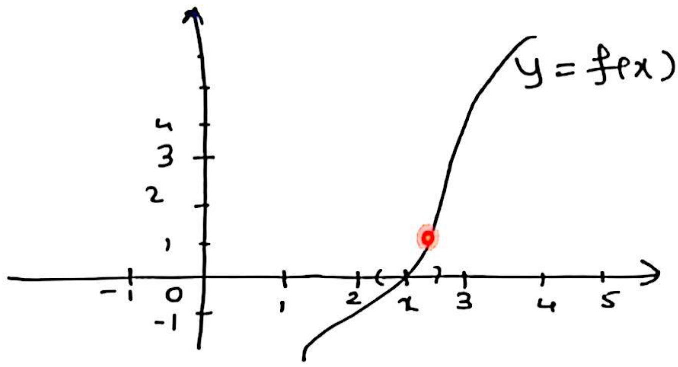
(5) Bisection method

How to choose initial approximation

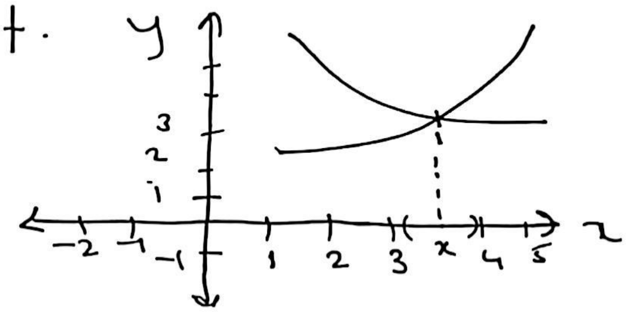
(1) By Intermediate value theorem :-
with the help of this theorem we can find the interval say $[a, b]$ in which $f(x)$ has root. we can choose any value between a & b as a initial approximation but best approximation is arithmatic mean of a & b

i.e. $x_0 = \frac{a+b}{2}$

(ii) we know that, root of $f(x)=0$ is the value of x at which graph of $y=f(x)$ intersects x-axis
"any value in the neighborhood of this point may be taken as an initial approximation to the root."



(iii) If function $f(x) = 0$ can be written in the form of $f_1(x) = f_2(x)$ then the point of intersection of $y = f_1(x)$ & $y = f_2(x)$ gives the root of $f(x) = 0$. therefore any value in the neighborhood of this point can be taken as initial approximation to the root.



ution of Algebraic & Transcedental Equation :-

<u>Bisection Method</u> (Bolzano method)

This method is based on the repeated application of intermediate value property

Let $f(x)$ be continuous between a & b

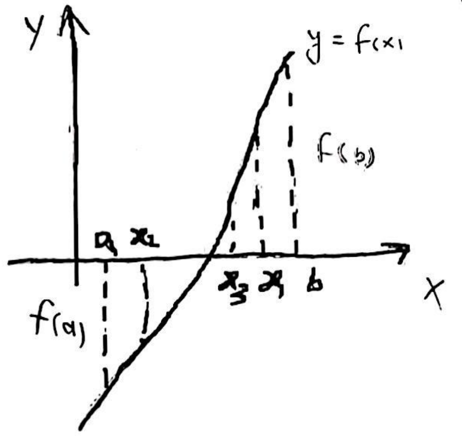
& let $f(a)$ be -ive,

& $f(b)$ be +ve

Then the first approximation of the root is $x_1 = \frac{a+b}{2}$, if $f(x_1) = 0$, then x_1 is root of $f(x) = 0$

Otherwise, the root lies between a & x_1 or x_1 & b, according as $f(x_1)$ is Positive or Negative

Then we bisect the interval as before & continue the process until the root is found accuracy



Bisection method :-

working Rule :-

1.) Let $f(x)=0$ — ①

be the given equation.

Find a and b such that

$f(a)<0$ and $f(b)>0$ (or $f(a)\cdot f(b)<0$)

2) Find first approximate root using Bisection method

$$x_1 = \frac{a+b}{2}$$

Calculate $f(x_1)$ and examine its sign.

2.1) If $f(x_1)<0 \Rightarrow$ root lies between x_1 and b. And 2nd approximate root is given by: $\boxed{x_2 = \frac{x_1+b}{2}}$

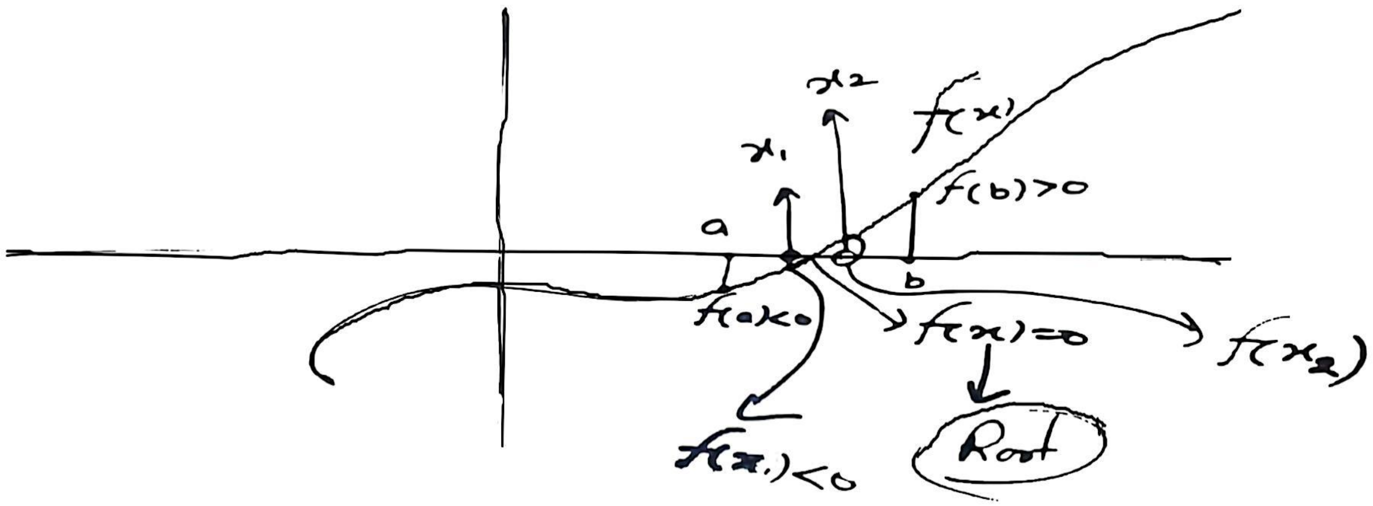
2.2) If $f(x_1)>0 \Rightarrow$ Root lies between a and x_1 then 2nd approximate root is given by

$$\boxed{x_2 = \frac{a+x_1}{2}}$$

Calculate $f(x_2)$ and Repeat step 2.1 and 2.2 until the required accuracy of root.

Concept of Bisection method

$f(x)=0$



Q: Find the real root of equation $x^3 - x - 4 = 0$ using Bisection method correct upto 3-decimal places

Sol: Let $f(x) = x^3 - x - 4 = 0$ ——— (1)

To Find a and b: $f(0) = -4 < 0$

$f(1) = 1 - 1 - 4 = -4 < 0$

$f(2) = 8 - 2 - 4 = 2 > 0$

$f(1.5) = -2.125 < 0$

$f(1.6) = -1.504 < 0$

$f(1.7) = -0.787 < 0$

$f(1.8) = 0.032 > 0$

Choosing $a = 1.7$ and $b = 1.8$

First approximate root using Bisection method

$$x_1 = \frac{a+b}{2} = \frac{1.7+1.8}{2} = 1.75$$

$f(1.75) = (1.75)^3 - 1.75 - 4 = -0.39 < 0$

Hence root lies b/w 1.75 and 1.8

2nd approximate root:-

$$x_2 = \frac{1.75+1.8}{2} = 1.775$$

$f(1.775) = (1.775)^3 - 1.775 - 4 = -0.182 < 0$

hence Root lies b/w 1.775 and 1.8

3rd approximate root:

$$x_3 = \frac{1.775+1.8}{2} = 1.7875$$

$f(1.7875) = (1.7875)^3 - 1.7875 - 4 = -0.076 < 0$

hence root lies b/w 1.7875 and 1.8

4th approximate root :-

$$x_4 = \frac{1.7875 + 1.8}{2} = 1.79375$$

$f(1.79375) = -0.022 < 0$

5th approximate root :-

$$x_5 = \frac{1.79375 + 1.8}{2} = 1.796875$$

$f(1.796875) = 0.0048 > 0$

hence root lies b/w 1.79375 and 1.796875.

6th approximate root :-

$$x_6 = \frac{1.79375 + 1.796875}{2} = 1.795312$$

$f(1.795312) = -0.0087 < 0$

7th approximate root

$$x_7 = \frac{1.795312 + 1.796875}{2}$$

$$\boxed{x_7 = 1.796093}$$

hence the approximate root correct upto 3-decimal places is $\boxed{x = 1.796}$